Questions by topic: Habitat case studies: woodland, rainforest, reefs, islands, mangroves, Antarctica - Set 1 (80 marks)

Answer all questions in the spaces provided.

0 1 . 1 Describe the ecological features of temperate broadleaf woodland and explain how woodland floor plants such as bluebells are adapted to them.

[4 marks]

0 1 . 2 Explain why an ancient woodland (existing before 1600) usually has higher biodiversity than a recently planted woodland.

[3 marks]

0 1 . 3 Describe **three** threats to UK broadleaf woodland.

[3 marks]

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0 2 Table 1 shows the number of bird species and butterfly species recorded in three woodlands in the same region.

Table 1

Woodland	Management	Bird species	Butterfly species
A	Ancient oak wood, coppiced on rotation with clearings	42	23
B	Ancient oak wood, unmanaged for 60 years, dense canopy	28	9
C	40-year-old conifer plantation	14	3

0 2 . 1 Use Table 1 and your knowledge to explain the differences in biodiversity between the three woodlands.

[5 marks]

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0 2 . 2 Describe **three** conservation efforts used to protect UK woodlands.

[3 marks]

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0 2 . 3 Explain why a high biodiversity increases the ecological stability of a woodland.

[2 marks]

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0 3 . 1 Explain why tropical rainforests have such high biodiversity.

[4 marks]

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0 3 . 2 Explain why rainforest species may be particularly vulnerable to environmental change.

[2 marks]

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0 3 . 3 Explain why rainforest plants and animals such as frogs are a rich source of medicines.

[2 marks]

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0 3 . 4 Explain the importance of tropical rainforest for the hydrological cycle and for soils.

[2 marks]

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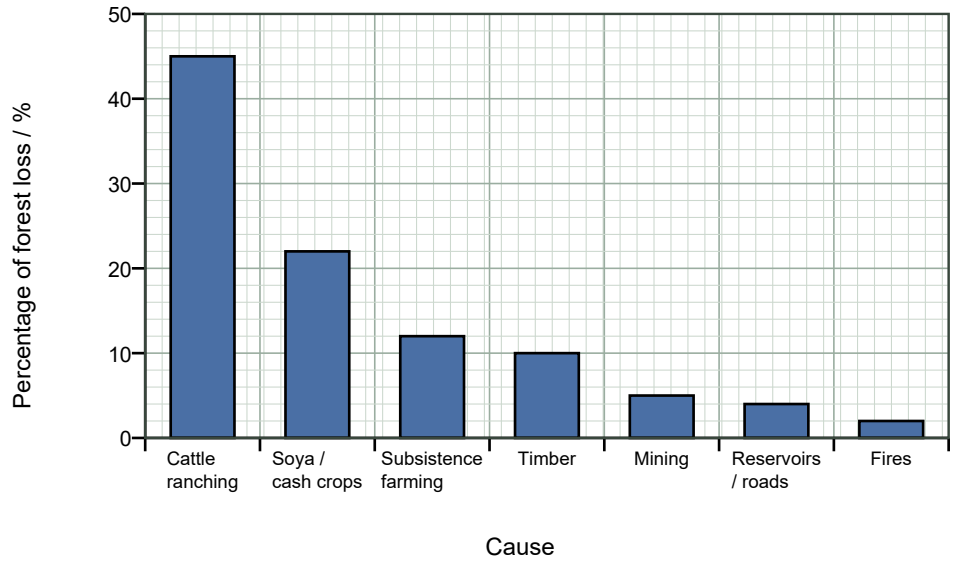
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0 4 Figure 1 shows the main causes of tropical rainforest loss in a South American country between 2000 and 2020.

Figure 1



0 4 . 1 Explain why traditional 'slash and burn' subsistence agriculture was sustainable but has become a threat.

[3 marks]

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0 4 . 2 Explain why clearance for cattle ranching or cash crops is usually unsustainable.

[2 marks]

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0 4 . 3 Describe how each of the following threatens rainforest.

[3 marks]

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Mineral extraction .....

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HEP reservoirs .....

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Climate change .....

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0 4 . 4 Describe how a debt-for-nature swap works.

[2 marks]

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0 5 . 1 Describe the conservation efforts used to protect tropical rainforests and evaluate their effectiveness.

[9 marks]

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0 5 . 2 State **one** reason why rainforest conservation is important to countries far from the tropics.

[1 mark]

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0 6 . 1 Explain how the relationship between coral polyps and zooxanthellae allows reefs to support a high biomass in nutrient-poor tropical seas.

[4 marks]

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0 6 . 2 Describe the abiotic conditions needed by reef-building corals and explain why these make reefs vulnerable.

[3 marks]

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0 6 . 3 Explain **three** ways in which coral reefs are important to humans.

[3 marks]

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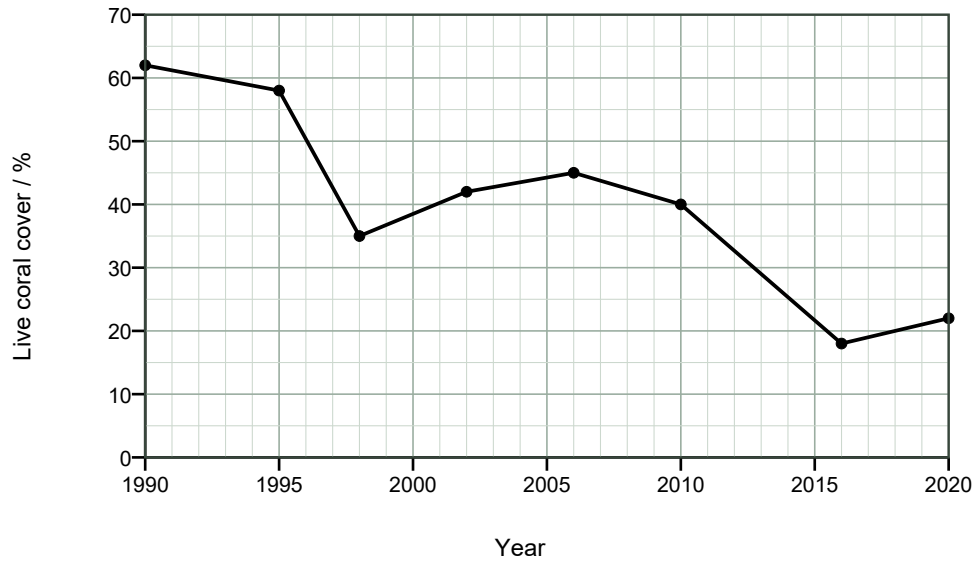
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0 7 Figure 1 shows the percentage of live coral cover on a reef near a growing coastal resort between 1990 and 2020, and the years of two major bleaching events.

Figure 1



0 7 . 1 Describe the changes in coral cover shown in Figure 1.

[2 marks]

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0 7 . 2 Explain what coral bleaching is and give **three** causes.

[4 marks]

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0 7 . 3 Explain how the growing resort could threaten the reef other than through bleaching.

[3 marks]

0 7 . 4 Explain why a no-take zone on the reef could increase fish catches in surrounding areas.

[1 mark]

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0 8 . 1 Explain how overfishing of the giant triton mollusc damaged coral reefs in Australia.

[2 marks]

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0 8 . 2 Describe **four** management practices used in marine protected areas to protect coral reefs.

[4 marks]

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0 8 . 3 Explain why deep-water coral reefs recover much more slowly from trawling damage than tropical reefs.

[2 marks]

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0 8 . 4 Give **two** threats to deep-water reefs and **one** conservation measure.

[2 marks]

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END OF QUESTIONS

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